

From Shell to Supercomputer: High Performance Workflows for Neuroimaging and Neurogenetics

Day 1 – Linux & Shell Fundamentals

Time	Session	Description	Type	Learning Outcome
09:00 – 10:00	Welcome & Orientation	Overview of CHPC, accessing systems, and workshop structure.	Learning	Understand the CHPC environment and secure access.
10:00 – 11:00	Command Line Basics	Navigation, files, permissions, and key Linux concepts.	Learning	Learn essential Linux commands and shell navigation.
11:00 – 11:15	Tea Break			
11:15-12:15	Guest Speakers: Dr Daniella Da Silva and Ms Caitlyn McMurchie Whitehead Scientific Pty (Ltd)			
12:15 – 13:15	Linux File Operations – Practice	Practice creating, moving, deleting, and searching files.	Hands-on	Apply Linux commands confidently to manage data.
13:15 – 14:15	Lunch Break			
14:15 – 15:15	Working with Text & Data	Use grep, cut, sort, awk, and pipelines for processing.	Hands-on	Manipulate data efficiently with text-processing tools.
15:15 – 16:00	Writing & Shell Scripts	Build scripts with variables, loops, and conditionals.	Hands-on	Write automation scripts and control execution flow.
16:00-16:30	Debugging Shell Scripts	Trace and fix issues using set -x, echo, and exit codes.	Hands-on	Debug scripts and interpret shell errors.

Day 2 – HPC Job Submission

Time	Session	Description	Type	Learning Outcome
09:00 – 10:00	HPC Overview	CHPC architecture, compute nodes, queues, and environment modules.	Learning	Understand HPC structure and terminology.
10:00 – 11:00	PBS Scheduler Concepts	Introduction to job scheduling and PBS script structure.	Learning	Learn the logic of PBS job configuration and submission.
11:00 – 11:15	Tea Break			
11:15 – 13:15	Writing PBS Scripts	Create and submit jobs using qsub, qstat, qdel.	Hands-on	Write and run job scripts on CHPC.
13:15 – 14:15	Lunch Break			
14:15 – 15:15	Monitoring & Debugging Jobs	Analyse logs, handle errors, and optimise jobs.	Hands-on	Diagnose job failures and refine resource requests.
15:15 – 16:00	Parallel Jobs & Arrays	Submit multi-subject jobs using PBS arrays	Hands-on	Execute batch and parallel processing workflows.
16:00- 16:30	Discussion & Review	Recap and group discussion.	Hands-on	Clarify uncertainties and consolidate HPC concepts.

Day 3 – Containers & AI in Scripting

Time	Session	Description	Type	Learning Outcome
09:00 – 10:00	Introduction to Containers	Why containers matter; pulling and running Singularity images.	Learning	Understand container use in HPC.
10:00 – 11:00	AI for Scripting	Overview of AI tools like ChatGPT, Copilot, and Cursor for research scripting.	Learning	Recognize how AI assists in writing and debugging scripts.
11:00 – 11:15	Tea Break			
11:15 – 12:45	Building a Singularity Container	Create a definition file and build a simple container.	Hands-on	Build and execute Singularity containers.
12:45 – 13:45	Lunch Break			
13:45 – 14:45	AI-assisted Scripting	Use AI tools to generate or optimize shell and PBS code.	Hands-on	Apply AI in practical scripting scenarios.
14:45 – 15:30	Integration Exercise	Combine scripting, job submission, and containers.	Hands-on	Integrate multiple skills in a reproducible workflow.
15:30 – 16:00	Wrap-up & Certificates	Feedback and certificate presentation.	Hands-on	Reflect on workshop outcomes and next steps.